



Cipangopaludina japonica

(Japanese mysterysnail)

Mollusks-Gastropods

Exotic

Collection Info

Point Map

Species Profile

Animated Map

Impacts



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Cipangopaludina japonica (von Martens, 1861)

Common name: Japanese mysterysnail

Synonyms and Other Names: Japanese mysterysnail, Oriental mysterysnail, Japanese black snail, Japanese trapdoor snail, *C. japonicus*, *C. chinensis japonica*, *C. chinensis japonicus*, *Bellamy* *japonica*, *B. chinensis japonica*, *Viviparus japonica*, *V. japonicus*, *V. chinensis japonica*, *Heterogen japonica*

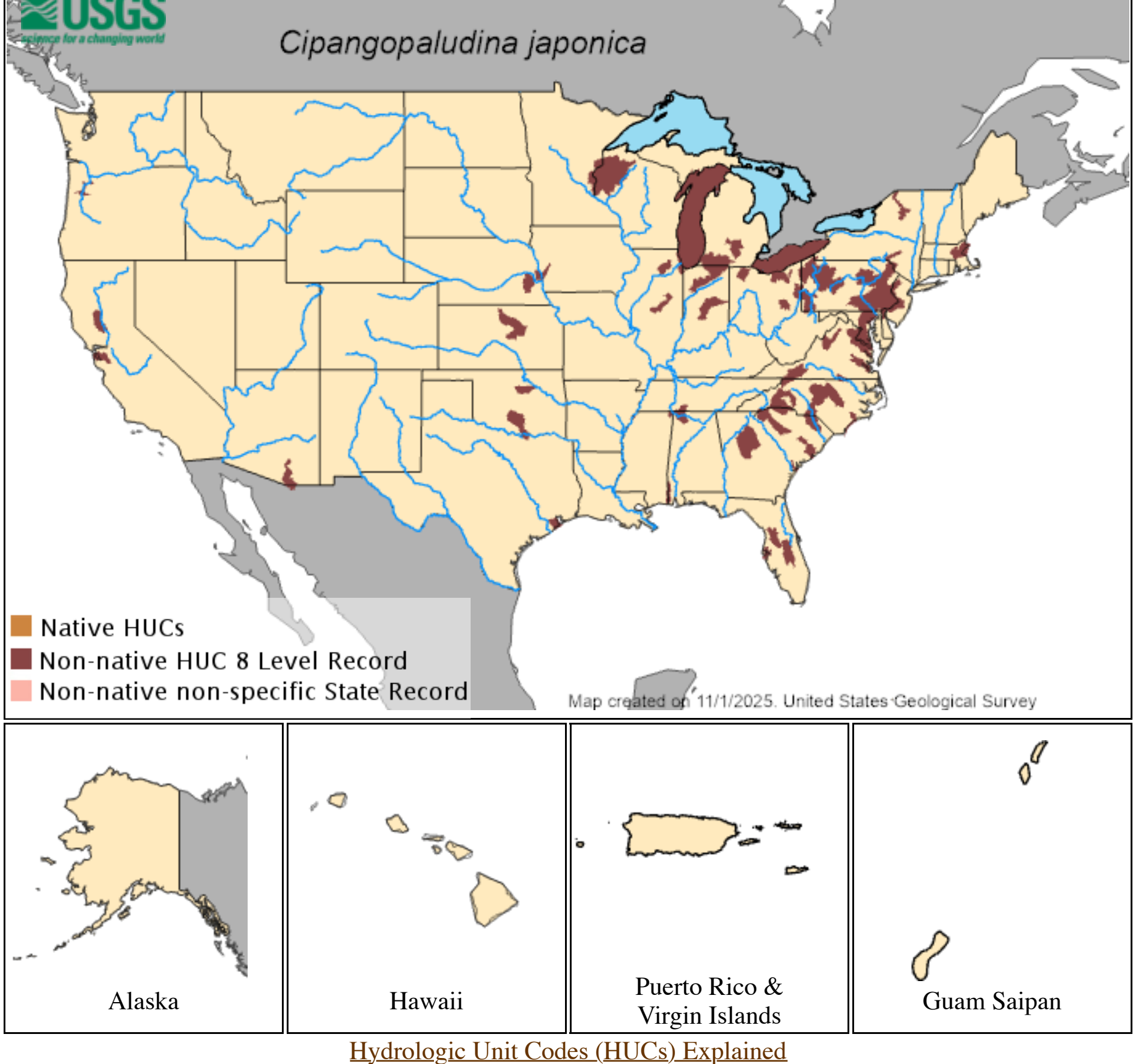
Taxonomy: available through 

Identification: The genus *Cipangopaludina* can be identified by its relatively large globose shells and concentrically marked opercula (Burch 1980). *Cipangopaludina japonica* exhibits a shell with 7–8 whorls, a very narrow umbilicus, and a spire that is produced at an angle of 50–55° (Jokinen 1992). Adult shells display fine carinae, while those of juveniles are covered in hairs on the periostracum where the carinae are located as well as around 8 striae/mm between the carinae in the middle of each whorl (Smith 2000). Individuals are light colored as juveniles and dark brown as adults (Wolfert and Hiltunen 1968).

The shell of *C. japonica* grows allometrically, the height increasing faster than the width, and does so at an increased rate in comparison with *C. chinensis*, so the adult shell is more elongate than that of the latter species (Jokinen 1982). The radula (feeding structure) also may differ between *C. japonica* and *C. chinensis*, but there is so much variation even within one species that it is not a good diagnostic characteristic (Smith 2000). However, in one North American population, *C. japonica* adults' radula had five small cusps on either side of the large central cusp and nine cusps on the marginal tooth (Jokinen 1982).

Size: can reach 50 mm. In Lake Erie, adults range from 30–65 mm high and 22–46 mm wide; females are slightly larger than males when mature (Wolfert and Hiltunen 1968).

Native Range: *Cipangopaludina japonica* is native to mesotrophic and eutrophic lakes in Japan (Jokinen 1992). Native to Japan, Taiwan, and Korea.



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Cipangopaludina japonica* are found [here](#).

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AL	1968	2003	2	Escatawpa: Pickwick Lake
AZ	1961	1961	1	Upper Santa Cruz
CA	1892	2022	2	Sacramento-Stone Corral: San Francisco Bay
DE	2013	2017	1	Brandywine-Christina
DC	2017	2018	1	Middle Potomac-Anacostia-Occoquan
FL	1949	1996	3	Kissimmee: Tampa Bay; Withlacoochee
GA	2013	2023	2	Upper Ocmulgee: Upper Oconee
IL	1995	2018	3	Chicago: Lower Fox: Upper Sangamon
IN	1965	2022	4	Ohio Region: St. Joseph: Tippecanoe: Upper White
KS	2005	2013	2	Lower Smoky Hill: Solomon
MD	2014	2023	3	Gunpowder-Patapsco: Middle Potomac-Anacostia-Occoquan: Patuxent
MA	1916	1968	2	Charles: Concord River
MI	1937	1965	2	Great Lakes Region: Upper Grand
MN	2007	2007	1	Lower St. Croix
MS	2007	2007	1	Lower Mississippi-Helena
NE	1998	2020	2	Big Papillion-Mosquito: Salt
NY	2016	2022	1	Raquette
NC	2004	2021	8	Deep: Haw: Lower Pee Dee: New River: Roanoke Rapids: Upper Catawba: Upper Neuse: Upper Pee Dee
OH	1945	2023	5	Black-Rocky: Cuyahoga: Lake Erie: Lower Maumee: Wills
OK	1953	1965	3	Lake Texoma: Lower Cimarron: Middle Washita
OR	2004	2004	1	Middle Willamette
PA	1962	2025	20	Brandywine-Christina: Clarion: Crosswicks-Neshaminy: French: Lackawaxen: Lehigh: Lower Delaware: Lower Monongahela: Lower Susquehanna: Lower Susquehanna-Swatara: Lower West Branch Susquehanna: Middle Allegheny-Redbank: Middle Allegheny-Tionesta: Middle Delaware-Musconetcong: Schuylkill: Shenango: Upper Delaware: Upper Ohio: Upper Susquehanna-Lackawanna: Upper West Branch Susquehanna
SC	1996	2025	7	Calibogue Sound-Wright River: Cooper: Lake Marion: Saluda: Seneca: Upper Broad: Upper Catawba
TX	2015	2015	1	West Galveston Bay
VA	2007	2023	6	Lower James: Lower Potomac: Lower Rappahannock: Middle Potomac-Anacostia-Occoquan: South Fork Shenandoah: Upper New
WI	2007	2016	4	Namekagon: Red Cedar: Upper Chippewa: Upper St. Croix

Table last updated 12/3/2025

† Populations may not be currently present.

Ecology: The Japanese mysterysnail is known to feed on detritus and sludge, both of which contain a myriad of different types of bacteria (Kurihara and Kadowaki 1988). In Japan, this species is commonly found in rice paddy fields on soil amongst higher plants (Kurihara and Kadowaki 1988).

Lives in freshwater rivers and lakes. The Japanese mystery snail in Sandusky Bay, Lake Erie, survives in conditions where surface water temperatures may reach 30°C, bottom water temperatures can reach 16–24°C, water is 4 m deep or less, there is high turbidity, the substrate is mud, and aquatic vegetation is sparse (Wolfert and Hiltunen 1968). It sometimes undergoes mortality events in marshes around Sandusky Bay in hot periods when waters dry up (Wolfert and Hiltunen 1968). In general in North America, the Japanese mystery snail has been found inhabiting waters of pH 6.3–7.3, calcium concentration of 11 ppm, sodium concentration of 16 ppm and conductivity of 62–194 µmhos/cm (Jokinen 1992).

This species is viviparous (Wolfert and Hiltunen 1968), giving birth to crawling young. Females live up to 8 years and are able to carry 10–120 young (Jokinen 1992). Young are generally born after water temperature rises to 15°C or more (Jokinen 1992).

Wolfert and Hiltunen (1968) found that the densities and distribution of the species in a waterbody is influenced by prevailing winds.

This species hosts many parasites, some of which affect human health (Michelson 1970; Lin and Chen 1980).

Means of Introduction: Imported into San Francisco at an Asian food market in 1892 (Wood 1892), and soon after found to be released into the San Francisco Bay, either from accidental disposal of shell or by purposeful release for food production (Wolfert and Hiltunen 1968). Deliberate release into the Great Lakes where it was intentionally stocked as food for the channel catfish, *Ictalurus punctatus* in Lake Erie in the 1940s (Mills et al. 1993, Wolfert and Hiltunen 1968).

Status: Established in many areas across the eastern US. This species is considered established in Lake Erie and reported from Lake Michigan.

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological

Economic

Human Health

Other



Recreation

This species reaches very high densities, and is considered a nuisance to fisherman; it has been caught in very large numbers by fishermen in Sandusky Bay, Lake Erie, in fact, two tons have sometimes been caught in one seine haul (Wolfert and Hiltunen 1968).

Parasite, Human Health

In the U.S., the Japanese mystery snail has been found to be a regular host to the common native parasite *Aspidogaster conchicola*, which is a first-time record in North America for a gastropod acting as host to this species (Michelson 1970).

The Japanese mysterysnail is a host to *Angiostrongylus cantonensis* larvae, a species associated with eosinophilic meningitis (Lin and Chen 1980).

Human Health, Agriculture

The ready ingestion of sewage sludge in rice fields by *Cipangopaludina japonica* has implications for wastewater treatment problems in urban areas (Kurihara and Kadowaki 1988).

Futher impacts

The Chinese mysterysnail (*Cipangopaludina chinensis*) is of the same genus as the Japanese mysterysnail; the species are difficult to distinguish from one another morphologically (see 'Remarks' section below; Jokinen 1982), and the Chinese mysterysnail appears to have been studied more heavily for its impacts, which can be viewed [here](#).

Remarks: Taxonomy of the introduced populations of Asian mysterysnails is confusing and there are many different scientific names in use. There has also been debate regarding whether or not *C. chinensis malleata* and *C. japonica* in North America are synonymous and simply different phenotypes of the same species. This database considers the two as separate species. Smith (2000) argues that *Cipangopaludina* is a subgenus of *Bellamy*; however, because most North American literature does not use the genus *Bellamy* to refer to these introduced snails, the mysterysnails discussed here are referred to by *Cipangopaludina*.

Heterogen japonica is used by Hirano et al (2019) based on mulple genetic markers.

David and Cote (2019) did a genetic and morphological analysis on North American populations of both *C. japonica* and *C. chinensis*, finding them genetically distinct, morphologically indistinguishable, and co-occurring in multiple lakes of New York; the authors go on to discuss literature which also supports the idea that these two species have no shell characters that can be used to distinguish them morphologically.

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Other Resources:
[Great Lakes Water Life](#)

[US Fish and Wildlife Service Ecological Risk Screening Summary for Cipangopaludina japonica](#)

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